

Lab 04 – Conceptual Design of an IIoT Sensor Network & Protocol Experimentation

Course Name: ITAI-3377

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Date: February 16, 2026

Introduction

This reflection delves into my experience in designing and simulating an Industrial Internet of Things (IIoT) sensor network aimed at real-time monitoring of temperature and humidity. The goal of this journal is to provide a critical assessment of the technical development process, the challenges faced, the knowledge acquired, and the wider real-world implications of the system I created. Although the project seemed like a straightforward coding task at first, it transformed into a more profound investigation of streaming data architectures and industrial monitoring systems.

Experience Overview

The laboratory project involved the development of a simulated IIoT sensor network designed to generate environmental data and visualize it in real-time through Python and Matplotlib animations. This system updates temperature and humidity readings every second and retains the latest 100 samples using deque data structures. These structures automatically discard older entries, mimicking the way actual industrial dashboards prioritize the most recent sensor data.

While the current implementation utilizes randomized values to simulate MQTT data transmission, its design embodies a genuine publish-subscribe model that is prevalent in IIoT settings. The visualization dynamically modifies axis limits and timestamps, emulating the

functionality of live monitoring dashboards found in manufacturing facilities, smart buildings, and environmental monitoring systems.

Personal Reflection and Critical Analysis

Initial Thoughts and Feelings

At the outset of this assignment, I viewed it as a simple technical endeavor centered on graphing data. However, as I worked on the animation loop and witnessed the ongoing updates, I started to grasp the intricacies involved in real-time systems. Observing the data in motion shifted my viewpoint from merely "writing code" to actually "designing a live monitoring system."

Initially, I found it challenging to comprehend how the animation function could continuously update the figure without causing the interface to freeze. Debugging the axis scaling and timestamp labeling necessitated careful consideration of how dynamic data should be presented without overwhelming the user. This challenge enhanced my understanding that visualization transcends mere number display; it is about conveying data in a clear and meaningful manner.

Challenges Faced

A major challenge encountered was more conceptual than technical: understanding that simulated data does not equate to actual sensor integration. While the program operated correctly, I identified a limitation in relying on randomized placeholder values instead of connecting to a genuine MQTT broker. This underscored the disparity between simulation and true IIoT implementation.

Another challenge was related to the real-time scaling of the x-axis and y-axis. Maintaining the readability of the graph during continuous updates necessitated modifications to tick intervals and axis limits. This experience emphasized the significance of usability in dashboard design.

What I Discovered

From this experience, I acquired a more profound insight into:

How streaming architectures manage ongoing data

The significance of effective data structures such as deques for handling constrained memory

How animation loops replicate real-time updates

The practical connection between sensors, communication protocols, and dashboards

Most crucially, I learned to adopt a systemic perspective. Instead of concentrating solely on Python syntax, I started to evaluate how each element sensor generation, protocol transmission, data buffering, and visualization integrates into a broader IIoT ecosystem.

Practical Implementation

This initiative is closely tied to industrial and smart-city uses. Real-time monitoring systems play a vital role in manufacturing facilities for identifying temperature irregularities, in agriculture for tracking humidity levels, and in smart buildings for optimizing energy use.

Gaining insight into how data streams are updated in real-time equips me for future roles that involve MQTT brokers, edge devices like Raspberry Pi, and authentic industrial sensors. If enhanced with actual protocol integration (MQTT, CoAP, or OPC UA), this system could serve as a streamlined industrial monitoring dashboard.

The project also highlights the importance of scalability. In practical applications, the volume of data can rise dramatically based on traffic patterns, production cycles, or changes in the environment. Creating systems that can effectively manage such fluctuations is essential in IIoT engineering.

Personal Growth and Future Improvements

This lab has significantly boosted my confidence in the design of real-time systems from an academic perspective. On a technical level, I have enhanced my skills in debugging dynamic functions and managing the visualization of streaming data. Conceptually, I have gained a deeper appreciation for architectural design within IoT environments.

Looking ahead, I plan to improve this system by:

- Integrating a live MQTT broker
- Comparing various protocols (MQTT vs. CoAP vs. OPC UA)
- Deploying the visualization on an edge device
- Implementing alert thresholds for anomaly detection

Conclusion

This lab shifted my perspective from basic data plotting to understanding the structure and scalability of real-time IIoT systems. The challenges I encountered, particularly around real-time visualization and system integration, deepened my analytical thinking. Beyond technical skills, I gained insight into how industrial monitoring systems function in real-world environments.

Reflecting on this experience has reinforced the importance of bridging theory with implementation. What began as a coding assignment became a meaningful exploration of IIoT architecture, preparing me for more advanced projects in edge computing and industrial automation.

Conclusion

This laboratory experience transformed my viewpoint from simple data visualization to comprehending the architecture and scalability of real-time Industrial Internet of Things (IIoT) systems. The obstacles I faced, especially regarding real-time data visualization and system integration, enhanced my analytical skills. In addition to technical expertise, I acquired a deeper understanding of how industrial monitoring systems operate in practical settings.

Contemplating this experience has emphasized the significance of connecting theoretical knowledge with practical application. What started as a programming task evolved into a significant investigation of IIoT architecture, equipping me for more complex endeavors in edge computing and industrial automation.